

Tiny Airline

Passenger Satisfaction Analysis

Turning passenger insights into a better travel experience

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Executive Summary

Problem

Only **43.45**% of passengers are satisfied, leaving 56.55% neutral or dissatisfied

Objective

Increase passenger satisfaction by **10%** within 6 months

Key Findings

- Online boarding satisfaction gap **1.37**
- In-flight entertainment gap **1.07**
- Departure - arrival delay correlation **0.96**

Recommendation

- Improve Digital Experience
- Strengthen In-flight Experience
- Improve Operational Reliability

Business Background & Dataset Overview

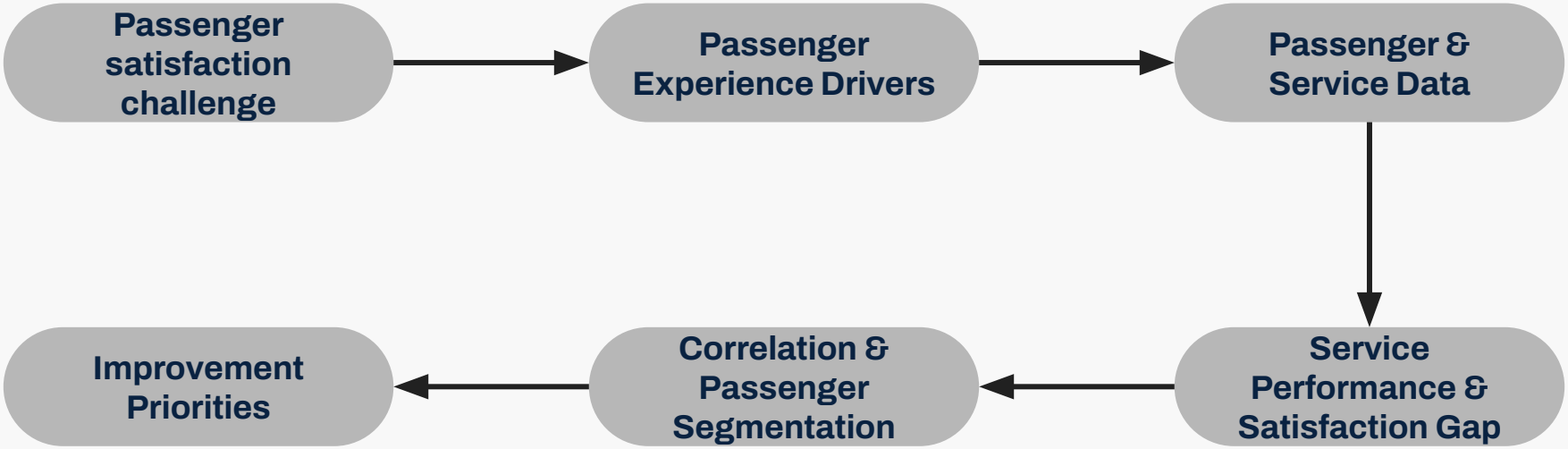
Business Background

- **Service Experience**
  Passenger satisfaction reflects the quality of services experienced throughout the journey
- **Digital Experience**
  Online booking and boarding are important digital touchpoints in the passenger journey
- **Operational Reliability**
  Departure and arrival performance contribute to the overall travel experience

Dataset Overview

129.880 **Passengers**	**24** **Variables**
14 **Service Attributes**	**1** **Satisfaction Outcome**

Methodology



Problem Statement

“How can Tiny Airline improve its passenger satisfaction rate by 10% within the next 6 months?”

Objective

The airline aims to **improve its passenger satisfaction rate by 10% within the next 6 months** by identifying and prioritizing the service and operational factors most strongly associated with passenger dissatisfaction using available passenger survey data.

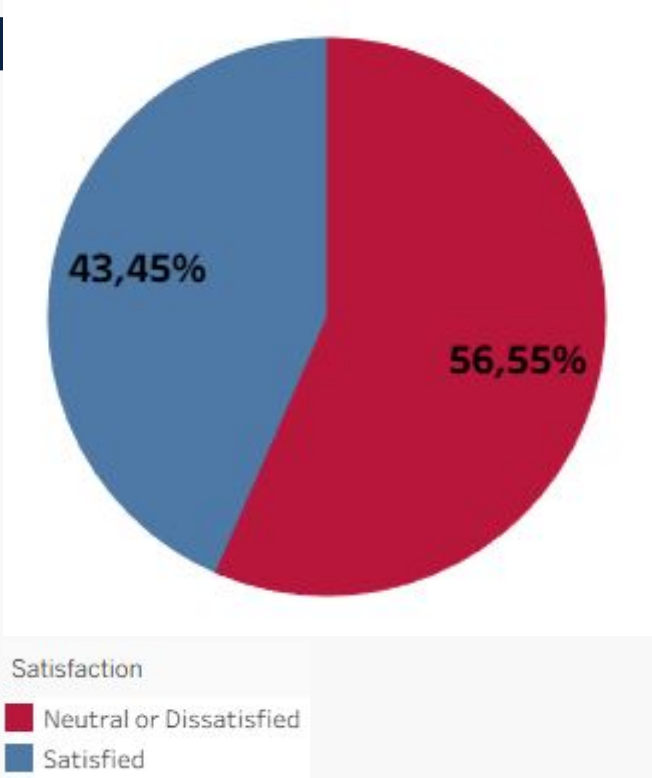
Hypothesis

Issue Branch	Possible Root Cause	Hypothesis	Prioritization	Metrics	Metrics Reasoning
Pre-flight & Digital Experience	Digital Journey	Lower online boarding and online booking ratings are more likely to report lower overall satisfaction	**High** - Digital touchpoints are part of the passenger journey and online boarding shows a substantial satisfaction gap	• Satisfaction Rate	Evaluates whether differences in digital experience are associated with overall passenger satisfaction
In-flight Experience	All In-flight Sub-groups	Lower ratings across In-flight experience attributes are associated with lower passenger satisfaction, while several attributes may also move together rather than independently	**High** - In-flight attributes show meaningful satisfaction differences, while several attributes are moderately to strongly correlated	• Average Service Rating • Satisfaction Rating Gap • Correlation Coefficient	Evaluates which In-flight attributes are associated with satisfaction and whether multiple attributes tend to move together
Operational Reliability	Departure Delay & Arrival Delay	Longer departure and arrival delays are associated with lower passenger satisfaction, while both delay measures tend to move closely together.	**High** - Operational delays can disrupt the passenger journey, while departure and arrival delay show a very strong relationship	• Average Arrival Delay • Correlation Coefficient	Evaluates whether delays are associated with satisfaction and whether departure disruption is closely related to arrival performance

Support Finding

Issue Branch	Possible Root Cause	Hypothesis	Prioritization	Metrics	Metrics Reasoning
Passenger Segmentation	Travel Profile	Passenger satisfaction may differ across class and type of travel	**High** - Class and type of travel already show meaningful differences in satisfaction and can help identify priority segment	• Satisfaction Rate by class • Satisfaction Rate by Type of travel	Identifies travel segments with the greatest improvement opportunity

MORE THAN HALF OF PASSENGERS ARE NOT YET SA



Only 43.45% of passengers were satisfied, while 56.55% were neutral or dissatisfied, showing a substantial opportunity to improve the overall passenger experience

Passenger Satisfaction Rating with Clear Gaps Across Service Experience



Finding: Only **43.45%** of passengers are **satisfied**, while **56.55%** are **neutral or dissatisfied**
Insight: Satisfaction differs substantially across service attributes

Major Opportunity to Improve Passenger Satisfaction





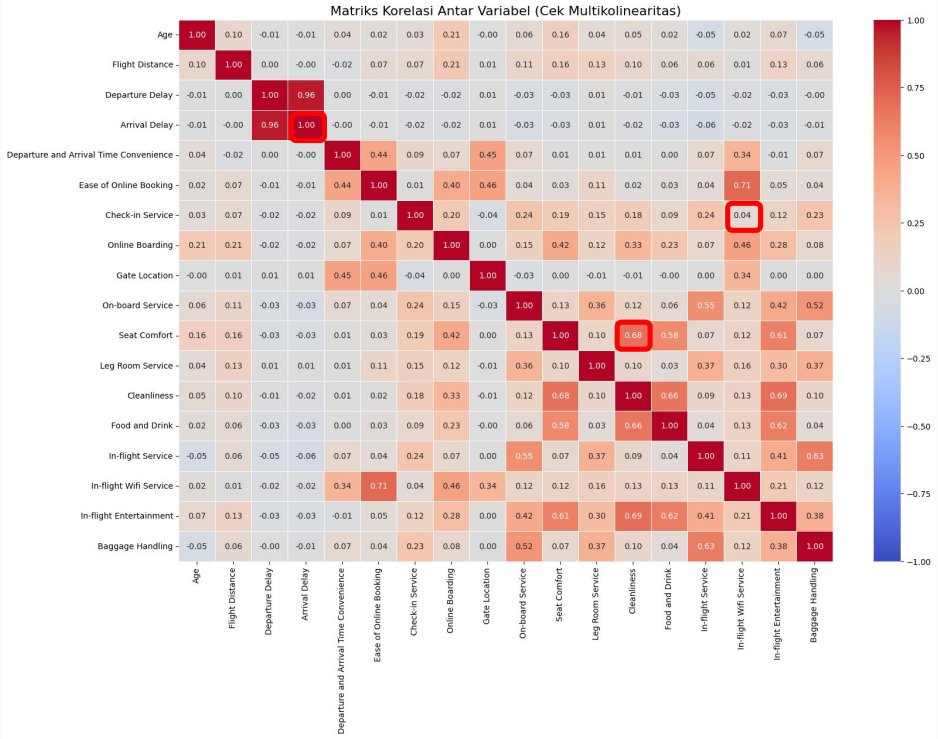
Top 3 satisfaction gaps



KEY FINDING

Online boarding shows the largest rating difference between satisfied and dissatisfied passengers.

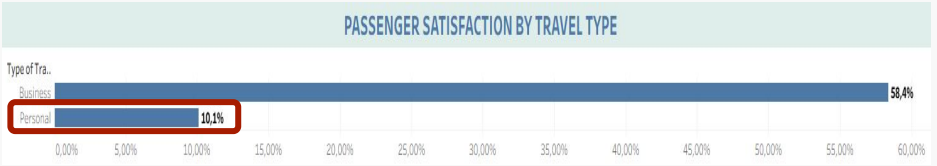
Multicollinearity



1. Departure and arrival delay show a very strong positive correlation (r = 0.96), indicating that both operational delay metrics tend to move closely together.
2. Ease of online booking and in-flight wifi show a strong positive correlation (r = 0.71), suggesting a related pattern in passengers' digital experience ratings.
3. Several in-flight experience attributes show moderate to strong correlations (r = 0.61–0.68), indicating that passengers' evaluations of these services tend to move together.
4. Several service and operational attributes are interconnected. Improvement opportunities should be focus on service areas that combine low performance and evaluated together when service attributes show strong relationships.

ECONOMY AND PERSONAL TRAVEL SHOW LOWER SATISFACTION





Insight	Business Alignment	Action
Economy passengers record the lowest satisfaction rate by class, while personal travel passengers show lower satisfaction than business travel passengers	These segments represent a larger improvement opportunity and can be prioritized when testing passenger experience initiatives	Pilot digital and in-flight service improvements among economy and personal travel passengers, then monitor whether their satisfaction rate improves after implementation

KEY INSIGHTS & BUSINESS RECOMMENDATION

Digital Experience

1.37
Online boarding
Satisfaction Gap

Largest service differentiation

Improve the end to end digital journey, particularly online booking and online boarding

In-flight Experience

1.07
Entertainment
Satisfaction Gap

Improve In-flight experience holistically, with priority on entertainment and passenger comfort

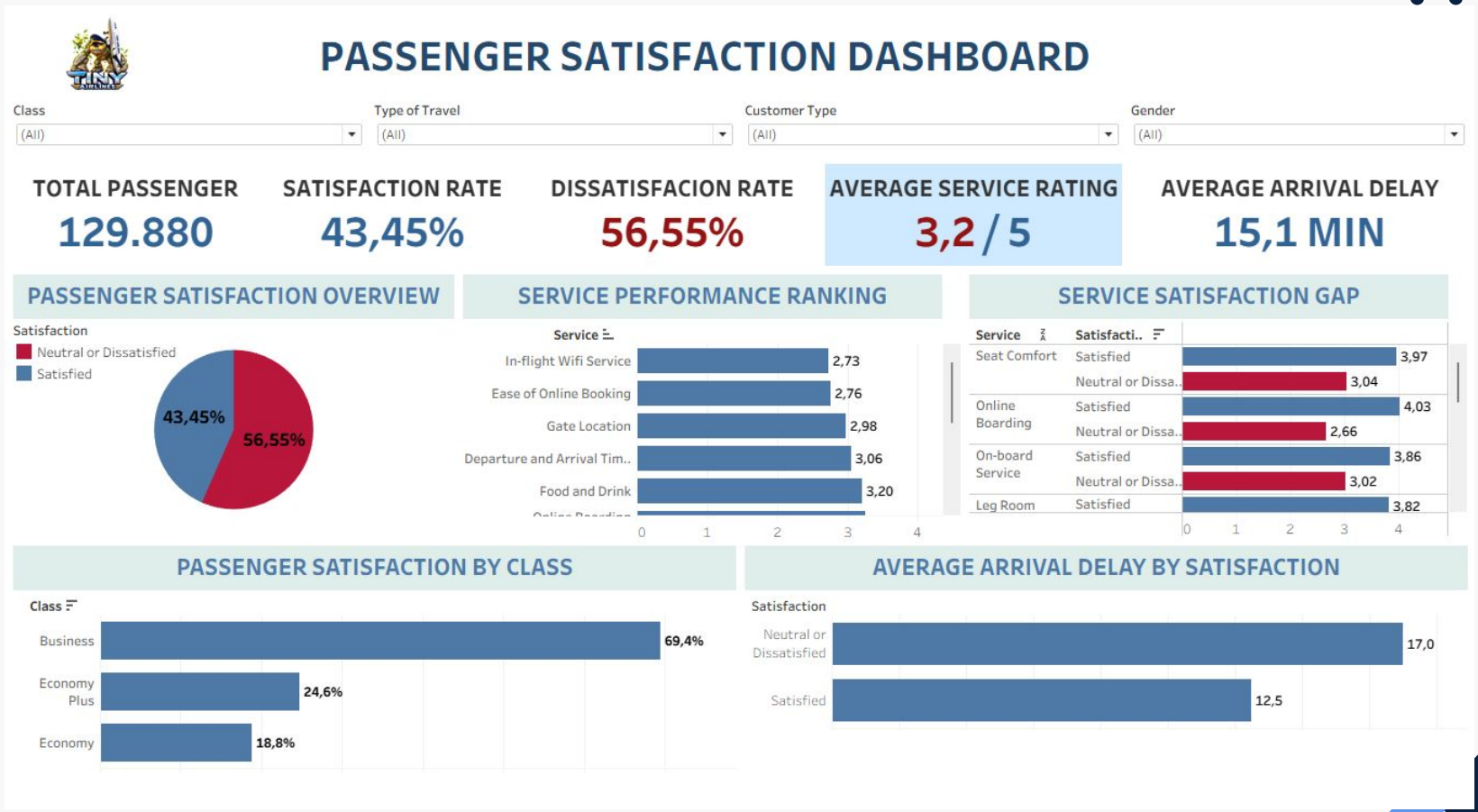
Operational Reliability

0.96
Departure - arrival
Delay Correlation

Delays move closely together

Improve departure punctuality and investigate recurring operational delay patterns

DASHBOARD


PASSENGER SATISFACTION DASHBOARD
Class
(All)
Type of Travel
(All)
Customer Type
(All)
Gender
(All)
TOTAL PASSENGER
129.880
SATISFACTION RATE
43,45%
DISSATISFACION RATE
56,55%
AVERAGE SERVICE RATING
3,2 / 5
AVERAGE ARRIVAL DELAY
15,1 MIN
PASSENGER SATISFACTION OVERVIEW
Satisfaction
Neutral or Dissatisfied
Satisfied
43,45%
56,55%
SERVICE PERFORMANCE RANKING
Service
In-flight Wifi Service 2,73
Ease of Online Booking 2,76
Gate Location 2,98
Departure and Arrival Tim.. 3,06
Food and Drink 3,20
0 1 2 3 4
SERVICE SATISFACTION GAP
Service
Satisfacti..
Seat Comfort
Satisfied 3,97
Neutral or Dissa.. 3,04
Online Boarding
Satisfied 4,03
Neutral or Dissa.. 2,66
On-board Service
Satisfied 3,86
Neutral or Dissa.. 3,02
Leg Room
Satisfied 3,82
0 1 2 3 4
PASSENGER SATISFACTION BY CLASS
Class
Business 69,4%
Economy Plus 24,6%
Economy 18,8%
AVERAGE ARRIVAL DELAY BY SATISFACTION
Satisfaction
Neutral or Dissatisfied 17,0
Satisfied 12,5

Thanks!

Appendix

- https://drive.google.com/drive/folders/10yWwunP-starkRcNudNPuFN2-HScTW94?usp=drive_link
- https://docs.google.com/presentation/d/19m3TuObvD8xVIF6GFXSzjRabiOCMSi97luZjlQmUC1Q/edit?usp=drive_link
- https://colab.research.google.com/drive/14lwMiXriZrNy0hYZPWCCo30qlIZ_hwAO?usp=drive_link
- https://drive.google.com/file/d/1TFcQs61Xd8emJjXlkh5dPLSJqmzfjcGY/view?usp=drive_link